OPHELIA Library
Reference manual



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Chapter 1

Introduction

The name "Ophelia" stands for "Optimization and Harmless Estimation Library for AMLET". The painting "Ophelia" in the front cover is due to Arthur Hughes in 1865. As reflected by its name, this library is part of the larger project AMLET, available at the address http://amlet.slashbin.net. This manual covers the version 0.7.1 of `Ophelia`.

1.1 Requirements

`Ophelia` relies heavily on the library `Oratio` (http://oration.slashbin.net), and the associated dependencies. In particular, `Ophelia`needs a proper `LAPACK`, as for `Oratio`, so we refer to `Oratio` documentation for more details. Operations requiring more advanced linear algebra, especially those present in `LAPACK`. While some `LAPACK` libraries are delivered with proper `LAPACK`implementations, they are often given standalone, so a copy of required `LAPACK` functions are embedded in `Ophelia`, and we will follow the associated conventions.

1.2 Conventions

$x(i)$ designs the i-th component of the vector x. When comparing two vectors x and y in $\mathbb{R}^n$, an inequality $x \leq y$ is meant componentwise, that is $x(i) \leq y(i),\ i = 1, \ldots, n$. We also follow the matrix conventions presented in the `NTools` component of `Oratio`.

Chapter 2

Basic operations

2.1 Matrix operations

Given an invertible symmetric matrix $A \in \mathcal{R}^{n \times n}$. the following function computes A^{-1}:

```
int op_matrix_sy_inverse(const enum CBLAS_ORDER Order,
                         const enum CBLAS_UPLO Uplo,
                         double *I, int n);
```

`Order` indicates if the matrix is stored by rows or columns, and take the constants `CblasRowMajor` and `CblasColMajor`, respectively. Since the matrix is supposed to be symmetric, only the lower or upper has to be filled, as indicated by the paramter `Uplo`, taking the value `CblasLower` or `CblasUpper`, respectively. The matrix itself is designed by I, and the dimension n is given as the last parameter. The routine overwrite the original with its inverse, storing only the upper or lower, following original storage.

Chapter 3

Unconstrained optimization

We consider the general problem

$$\min_{x \in \mathcal{R}^n} f(x). \tag{3.1}$$

3.1 Linesearch methods

BFGS Linesearch

3.2 Trust-region methods

3.2.1 Unconstrained optimization

3.3 Hessian approximations

While the gradient $\nabla f(x)$ usually must be known accurately in minimization algorithms, we often can rely on approximations of the Hessian $\nabla^2 f(x)$. This is especially appealing since finite or central differences require $O(n^2)$ function evaluations, which is prohibitive in many applications. In this section, we will focus on traditional secant updates, by focusing on their definitions, but not on the underlying convergence theory. We refer the reader interested in such issues to Chapter 8 of Dennis and Schnabel [2].

Reefering to the original problem 3.1, suppose that we have just taken a step from x_c to x_+. Following Dennis and Schnabel, we consider the Hessian as the derivative matrix of the nonlinear equations system $\nabla f(x) = 0$. The secant equation for the unconstrained minimization problem is

$$H_+ s_c = y_c, \tag{3.2}$$

where

$$s_c = x_+ - x_c,$$
$$y_c = \nabla f(x_+) - \nabla f(x_c).$$

Since the Hessian of some function is symmetric, we seek an H_+ that satisfies $H_+ = H_+^T$ as well as (3.2). The first secant update successfully used in the litterature is the symmetric secand update

BFGS

A popular approximation is the BFGS (Broyden, Fletcher, Goldfarb and Shanno):

$$B_{k+1} = B_k + \frac{y_k y_k^T}{y_k^t d_k} + \frac{B_k d_k (B_k d_k)^T}{d_k^T B_k d_k},$$

where $y_k = \nabla_x f(x_{k+1}) - \nabla_x f(x_k)$.

SR1

3.3.1 Trust-region radius update

Initial trust-region radius

Various options again exist to choose the initial trust-region as well as to update it.

The initialization strategy can be defined by setting the `radius_init` element of the `BTR` structure to the desired user's function, or to NULL, if no initialization has to be performed. In this case, the initial trust-region radius is affected to `BTR_INITIAL_DELTA`, defined in the file `defaults.h`.

The default behaviour of the algorithm sets the initial trust-region to $\|\nabla f(x_0)\|$, as described by Lin and More [4], while more advanced strategies exist. On the first iteration, the predicted step is used to compute a more reasonable trust-region radius, using the formula:

$$\Delta := \min\{\Delta, \|s_0\|\}.$$

Chapter 4

Constrained optimisation

In this chapter, we consider the problem

$$\min_{x \in \mathcal{C}} f(x),$$

where $\mathcal{C}$ is the feasible region. Usually, (C) is expressed as a collection of equalities and inequalities constraints:

$$\begin{aligned} c_i(x) &\leq 0, \ i = 1, \ldots, m_1, \\ c_i(x) &= 0, \ i = m_1 + 1, \ldots, m. \end{aligned}$$

4.1 Projection methods

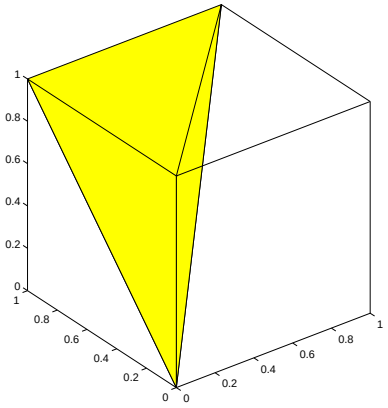
We will restrict here to the case where the feasible region C is a nonempty closed convex subset of $\mathbb{R}^n$, on which projections are relatively easy to calculate.

4.1.1 Bounds constraints

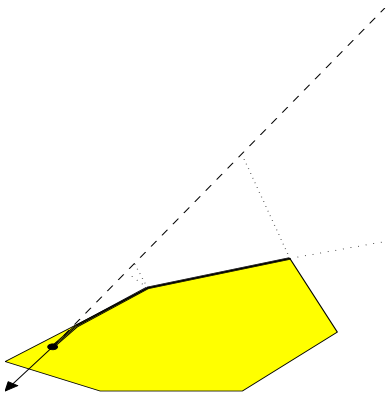
One of the simplest feasible regions $\mathcal{C}$ is obtained when problem's variables are subject to bounds constraints, i.e.

$$\mathcal{C} = \{x_\ell \leq x \leq x_u\}.$$

The vectors x_ℓ and x_u are the vectors of lower and upper bounds on the variables, respectively, and we allow either of them to have components $-\infty$ of $+\infty$.



the order simplex in 3D



the projected path

4.2 Stopping criteria

4.2.1 Unconstrained optimisation

Relative gradient. Dennis and Schnabel [2], Page 160.

$$\max_i \frac{\nabla f(x)_i \max\{|x_i|, typ\ x_i\}}{\max\{|f(x)|, typ\ f\}},$$

where x_i is the ith component of the vector x.

```
double nlp_relative_gradient(double val, double *x, double *grad,
                             double *typx, int n, double typf)
```

4.2.2 Constrained optimisation

Projection methods for trust-region methods

We use the criterion presented in Conn et al. [1], Section 12.1:

$$\|P[x - \nabla f(x)] - x\|_2 < \epsilon.$$

```
int btr_constrained_projection_stopping_test(BTR *btr, double rho)
```

$$\max_i \frac{(P[x - \nabla f(x)] - x)_i \max\{|x_i|, typ\ x_i\}}{\max\{|f(x)|, f(typ\ f\}},$$

where x_i is the ith component of the vector x.

4.2.3 Specific projection algorithms

Isotonic regression

Isotonic regression (IR) involves finding a weighted least-squares fit $x \in \mathbb{R}^n$ to a vector $a \in \mathbb{R}^n$ with weights vector $w \in \mathbb{R}^n$ subject to a set of monotonicity constraints giving a simple or partial order over the variables.

Overview

Chapter 5

Metaheuristics methods

While some metaheuristics algorithms are proposed, it should be stressed that the library does not focus on such methods, and their support is therefore very limited.

5.1 Simulated annealing

```
NlpSa *nlp_sa_new();
```

Chapter 6

Test problems

6.1 Unconstrained programs

6.1.1 Univariate functions

6.2 Parabola

$$x^2 + 4x - 10$$

Figure 6.1: `parabola_1` function

6.2.1 Multivariate functions

6.3 Bound constrained programs

6.3.1 Univariate functions

6.3.2 Multivariate functions

Hock

The following functions are taken from [3].

$$f(x, y) = \frac{(x+1)^3}{3} + y$$

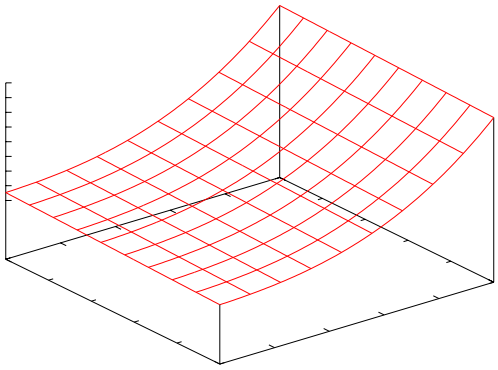


Figure 6.2: `pbr t1 3` function

6.4 Global optimization

A set of functions are provided to test the capabilities to escape from local minimizer and identify a global minimizer. These functions are characterized with the presence of several local minimizers, from where it is difficult for an algorithm to escape.

6.4.1 Univariate functions

GSL E1

This function has been taken from the GNU Scientific Library (http://www.gnu.org/software/gsl). The function is considered to test the simulated annealing algorithm, under the name `E1`, and has the form

$$f(x) = e^{-(x-1)^2} sin(8x) - 0.89e^{-(x-1000)^2}$$

Its graph is given in Figure 6.3, and its definition is

```
double gsl_e1(double x)
```

where x is the point where we want to evaluate the function. We denote if by `gsl e1`. The optimal value is $f^* = -0.87287$, attained at $x^* = 1.36312999$. The function has also a local minimizer at $x = 0.60146196$, for an objective value $f = -0.84893$.

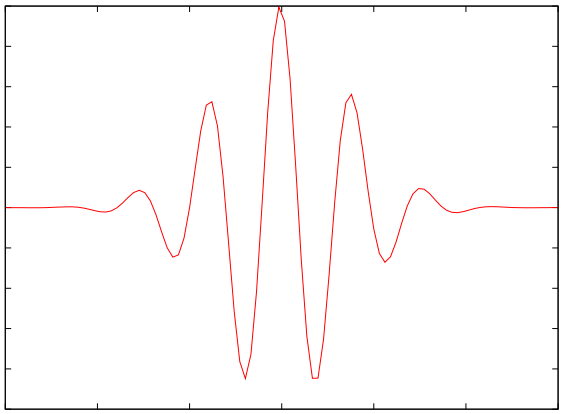


Figure 6.3: `gsl e1` function

6.4.2 Multivariate functions

Ackley's function

$$f(x,y) = -20e^{-0.2\sqrt{0.5(x^2+y^2)}} - e^{0.5(\cos(2\pi x)+\cos(2\pi y))} + e + 20 \qquad (6.1)$$

The global minimizer is $(0,0)$, with the corresponding objective value $f(0,0) = 0$.

Appendix A

Basic analysis

A.1 Banach spaces

A metric space is an ordered pair (M, d) where M is a set and d is a metric on M, that is, a function

$$d : M \times M \to \mathbb{R}$$

such that, for any x, y and z in M,

1. $d(x, y) \geq 0$;
2. $d(x, y) = 0$ if and only if $x = y$;
3. $d(x, y) = d(y, x)$;
4. $d(x, z) \leq d(x, y) + d(y, z)$.

The first condition follows from the other three, since:

$$2d(x, y) = d(x, y) + d(y, x) \geq d(x, x) = 0.$$

A more compact but equivalent definition replaces the above four axioms by the following two:

1. $d(x, y) = 0$ if and only if $x = y$;
2. $d(x, z) \leq d(x, y) + d(z, y)$.

The function d is also called distance function or simply distance. Often d is omitted and one just writes M for a metric space if it is clear from the context what metric is used.

A metric space M is said to be complete (or Cauchy) if every Cauchy sequence of points in M has a limit that is also in M or alternatively if every Cauchy sequence in M converges in M.

A seminormed vector space is a pair (V, p) where V is a vector space and p a seminorm on V. A normed vector space is a pair $(V, \|\cdot\|)$ where V is a vector space and $\|\cdot\|$ a norm on V. We often omit p or $\|\cdot\|$ and just write V for a space if it is clear from the context what (semi)norm we are using.

A Banach space is complete normed vector space. The familiar Euclidean spaces $\mathbb{R}^n$ and $\mathbb{C}^n$, where the Euclidean norm of $x = (x_1, \ldots, x_n)$ is given by

$$\|x\| = \sqrt{\sum_{i=1}^{n} x_i^2},$$

are Banach spaces.

A.2 Topological spaces

A topological space is a set X together with T, a collection of subsets of X, satisfying the following axioms:

1. The empty set and X are in T.
2. The union of any collection of sets in T is also in T.
3. The intersection of any finite collection of sets in T is also in T.

A topological space is called separable if it contains a countable dense subset; that is, there exists a sequence $\left\{_{x_n}\right\}_{n=1}^{\infty}\}$ of elements of the space such that every nonempty open subset of the space contains at least one element of the sequence.

Appendix B

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